Multiobjective Guided Divide-and-Conquer Network for Hyperspectral Pansharpening

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***Abstract*— Deep learning methods have gained rapid development in hyperspectral pansharpening (HP) due to powerful spatial–spectral feature extraction ability. However, most of these methods are optimized using a single reconstruction objective. It is difficult for these methods to find a balance between spectral preservation and spatial preservation. Furthermore, these methods adopt interpolation or convolution to upsample the hyperspectral images (HSIs), which tends to cause noticeable spectral distortion. To conquer these issues, a novel multiobjective guided divide-and-conquer network (MO-DCN) is proposed for HP. It consists of a deconvolution long short-term memories (LSTMs) network (DLSTM) and a divide-and-conquer network (DCN). DLSTM leverages bi-direction learning to upsample HSIs by considering 3-D spatiotemporal dependencies. Then, DCN designs a two-branch architecture to reconstruct spatial and spectral information from upsampled HSIs and panchromatic images (PANIs), respectively, where the spatial branch designs an attention-in-attention module (AIAM) to emphasize complementary attention in a coarse-to-fine way. Finally, co-improvement of spatial and spectral information is formulated as an Epsilon-constraint-based multiobjective optimization. The Epsilon constraint method transforms one objective into a constraint and regards it as a penalty bound to make an excellent tradeoff between different objectives. Experimental results demonstrated that the proposed method markedly improves pansharpening performance in both the spatial and spectral domains and has superior fusion performance than state-of-the-art methods.**

***Index Terms*— Deep learning, hyperspectral image (HSI), hyperspectral pansharpening (HP), multiobjective optimization, spatial–spectral fusion.**

I. Introduction

HYPERSPECTRAL sensors can record spectral and spatial information of the same scene simultaneously, which is widely used in surveillance [1], geoponics [2], mineralogy [3], military [4], astronomy [5], and so on [6]. Hyperspectral images (HSIs) are collected in the wavelength range from ultraviolet to mid-infrared spectrum. In the HSIs, the spectral resolution is growing quickly with the boost of the imaging spectrometer. Unfortunately, limited by the imaging hardware, the spatial resolution of the HSIs is sacrificed to ensure higher spectral resolution, which could not satisfy the requirements of some practical applications.

The panchromatic image (PANI) is a kind of single-band black and white image, which absorbs more luminous energy from the entire visible light to generate a higher spatial resolution image. Considering the complementary advantages of HSIs and PANIs, researchers try to use hyperspectral pansharpening (HP) technologies to fuse these two kinds of images. HP refers to recover high-resolution HSIs (HR-HSIs) from low-resolution HSIs (LR-HSIs) and high-resolution PANIs. It combines the superiority of discriminative spectral information of LR-HSIs and abundant spatial details of PANIs to complement each other. HP is an extension of multispectral pansharpening that fuses multispectral images and PANIs. Compared with multispectral pansharpening, HP is more difficult in spatial and spectral preservation because of lower spatial resolution and higher spectral dimensionality.

So far, traditional pansharpening methods have achieved vigorous development that can be mainly divided into four branches: component substitution (CS), multiresolution analysis (MRA), matrix factorization, and Bayesian methods. As a kind of widely used pansharpening methods, the CS-based methods decompose the 3-D data structure of LR-HSIs along with spectral and spatial components, and then substitute the spatial component with PANIs. Finally, an inverse transform is applied to reconstruct the HR-HSI. These kinds of methods are easy to implement and able to improve the spatial resolution effectively but easily suffer from noticeable spectral distortion. Compared with the CS-based methods, MRA families are superior in spectral preservation by injecting the spatial details of PANI to original LR-HSI directly. However, the extraction of spatial details may cause spatial distortion in most cases, such as ringing artifacts. The matrix-decomposition-based methods take the pansharpening as a factorization problem and typically use non-negative matrix factorization to generate HR-HSI with the priors of LR-HSI unmixing. The Bayesian methods implement the fusion process by constructing a Bayesian inference framework with a hypothetical prior

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distribution. These two branches are known as model-based methods, which deeply rely on prior information.

In the past decade, deep learning has received widespread attention because of its automatic feature extraction ability without prior knowledge. The deep-learning-based methods have been emerged and developed for HP in recent years, such as convolutional neural network (CNN) [7]–[14] and generative adversarial network (GAN) [16], [17]. There are two crucial issues in these deep-learning-based methods to solve HP tasks, architecture design, and optimization objectives' design. The architecture that upsamples the LR-HSIs first and then fuses the upsampled LR-HSIs with PANIs is widely used in HP. Most methods adopt fixed interpolation [8]–[11] or convolution operation [7], [12], [13] to upsample the HSIs, which can provide spatial super-resolution, but do not emphasize the reconstruction among long-term spectral bands. In terms of optimization objectives, the existing deep learning networks often use a single objective function to achieve spatial and spectral preservation. The single objective function is often designed in two forms. One is a single objective of independent terms, such as mean square error (MSE) and L1 loss. The other is a single objective of multiple mixed terms, where the adjustment factor is used to balance multiple terms. In HP, the spatial and spectral reconstructions are sometimes conflicting, and the difficulty of the reconstruction is different. The former tends to prefer the easy-to-optimize one in spatial and spectral reconstructions, which causes the risk of deterioration on the other one. For the latter, it is easy to be too large or too small for the manually adjusted parameter, which may deteriorate the spectral information or lose the spatial details. Thus, it is still a challenging problem for HP to find an excellent compromise between spectral preservation and spatial preservation during the optimization procedure.

In this article, a novel multiobjective guided divide-and-conquer network (MO-DCN) is proposed for HP. In MO-DCN, LR-HSI is first upsampled through a deconvolution long short-term memory (LSTM) (DLSTM) network. DLSTM constructs a bidirectional structure to learn long-term spatiotemporal dependencies from previous and following spectral bands. Afterward, the upsampled HSI and PANI are fused in a divide-and-conquer network (DCN). DCN designs dual branches to improve the spatial and spectral resolution of fused images, respectively. In the spatial branch, an attention-in-attention module (AIAM) is designed to aggregate the attention from coarse to fine by emphasizing discriminative spatial features and fusing complementary spatial information adaptively. The optimization of MO-DCN is formulated as a multiobjective optimization problem. To solve this problem, a new Epsilon-constraint-based method is devised to optimize two objectives of a spatial term and a constrained spectral term alternately. In the spectral term, the spatial term is transformed into a constraint under a bound penalty. Finally, MO-DCN is optimized in an end-to-end way to achieve spatial and spectral preservation simultaneously. The main contributions of this article can be summarized as follows.

1) In HP, the spatial–spectral reconstruction of two conflicting objectives is modeled as a new multiobjective optimization, which promotes MO-DCN to seek the excellent tradeoff Pareto-optimal solutions and reach a co-improvement in spatial and spectral preservation.
2) MO-DCN designs a divide-and-conquer strategy to construct specific spatial and spectral fusion architecture, which is able to adapt to various characteristics of spatial and spectral reconstructions. In the spatial reconstruction, the AIAM can not only extract the rich spatial information in PANI but also exploit the complementary spatial information from LR-HSI.
3) In MO-DCN, DLSTM is a customized upsampling network for LR-HSI, which can restore the spatial details as much as possible and alleviate the spectral distortion significantly. It provides the potential to improve spectral and spatial preservation.

The remainder of this article is divided into four sections. The related works will be presented in Section II. Afterward, Section III will introduce the overall structure of the proposed MO-DCN thoroughly. In Section IV, the detailed qualitative and quantitative experimental results and analysis are discussed to verify the effectiveness of the proposed algorithm. Finally, some conclusions and further work are given in Section V.

II. Related Work

In this section, we investigate traditional and deep-learning-based HP methods. The traditional HP methods mainly rely on specific mathematic models or filter designs to extract handcraft features for fusion. The deep-learning-based methods automatically extract hierarchical features via learnable components, such as convolution and full connection. The high-level abstract features enhance the robustness and generalization of the model greatly, which usually brings better spatial and spectral preservation. The details of the state-of-the-art HP methods are introduced as follows.

A. Traditional Methods

Some representative methods of four branches in Section I, CS [18]–[21], MRA [22]–[27], matrix factorization [28]–[30], and Bayesian [31]–[35], are presented. The CS-based methods have superiority in rendering spatial information, while the MRA-based methods perform better in spectral preservation. The representative CS-based methods include Gram–Schmidt (GS) [20], intensity–hue–saturation (IHS) [21], and adaptive GS (GSA) [19], to name a few. The representative MRA-based methods include generalized Laplacian pyramid modulation transfer function (MTF-GLP) [25] and smoothing filter-based intensity modulation (SFIM) [26]. Later, numerous variants of MTF-GLP are developed, such as MTF-GLP with high-pass modulation (MTF-GLP-HPM) [23].

The third branch is matrix factorization. Matrix factorization implements the HP under the guidance of the unmixing for LR-HSIs. Coupled non-negative matrix factorization (CNMF) [29] and the Lanaras method [30] predict the endmember matrix and abundance matrix in the unmixing, which are multiplied to generate the HR-HSI. The Bayesian methods construct an inference framework to regularize the fusion problem by assuming an appropriate prior distribution. Hysure [32],

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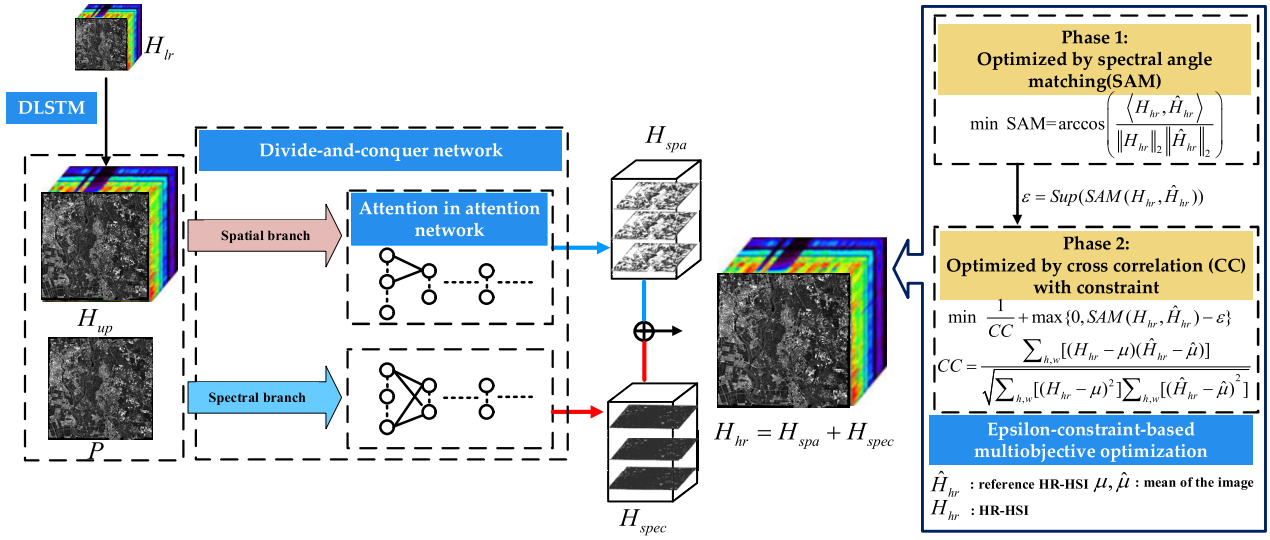


Fig. 1. Flowchart of the proposed MO-DCN.

Bayesian naive Gaussian prior [33], and Bayesian sparsity promoted Gaussian prior [34] belong to this category. Compared with the CS-based and MRA-based methods, matrix factorization and Bayesian methods require some specific priors or distribution assumptions.

B. Deep-Learning-Based Methods

The deep-learning-based HP methods mainly contain three parts, LR-HSI upsampling, spatial–spectral reconstruction, and optimization. Xie *et al.* [13] proposed a superresolution deep learning (SRDL) model to upsample the LR-HSI using several convolutional layers. Li *et al.* [7] upsampled the LR-HSI by successively conducting a Laplacian pyramid super-resolution network (LapSRN) in a band-by-band way, where the network uses transposed convolution and multiscale information fusion.

Afterward, a serial of joint spatial–spectral HP networks are devised and trained with different optimization objectives. The single objective with an independent term, such as MSE and L1 loss, is commonly used as a comprehensive learning objective. Zheng *et al.* [12] constructed a dual attention residual network trained by the L1 loss function, where spatial–spectral attention is merged into every residual block to extract deep discriminative features. He *et al.* [11] proposed a spectral constrained adversarial autoencoder. In the adversarial autoencoder, a distance loss in spectral angle is constrained for the latent feature space to mine deeper features.

HP requires both spatial and spectral preservation. These two requirements are usually conflicting. To alleviate the conflict between different requirements, multiple objectives are weighted summed to form a single objective. In [17], a 3-DGAN is proposed, where a 3-D spectral–spatial generator network is designed to upsample the LR-HSI and a discriminator network is designed to learn the spatial information preservation capability of PANI and LR-HSI, where MSE and the spectral angle matching (SAM) are combined as the objective function with a fixed coefficient. Dong *et al.* [16] devised a Gaussian–Laplacian pyramid deep convolutional network to learn different levels of details and define the fixed weight summation of L1 and correlation coefficient (CC) as the objective function. The conflicting objectives make the network prone to bias for spatial or spectral preservation. It is difficult for fixed weight summation to achieve a good balance between spatial and spectral preservation.

III. Proposed MO-DCN Method

The HP task is solved by a newly proposed MO-DCN framework, and MO-DCN consists of three parts: bidirectional and long-term learning-based upsampling, divide-and-conquer-based spatial–spectral fusion, and Epsilon-constraint-based multiobjective optimization. Fig. 1 shows the overall framework of the proposed MO-DCN method. In the first stage, the LR-HSI is upsampled to the same spatial scale as that of PANI by DLSTM, where bidirectional and long-term spatiotemporal features are learned. Considering different characteristics of spatial and spectral reconstructions, DCN is constructed to reconstruct spatial and spectral information, respectively. Then, HR-HSI is obtained by fusing the spatial and spectral information from these two branches. In the final stage, the whole architecture is optimized by a novel optimization method, Epsilon-constraint-based multiobjective optimization. It urges the network to find Pareto-optimal solutions and make a satisfying balance between spatial and spectral reconstructions.

A. Problem Formulation

Let the LR-HSI with $M \times N$ pixels and B-bands be denoted as $H_{\text{lr}} \in R^{M \times N \times B}$. The corresponding PANI is $P \in R^{\beta M \times \beta N}$, and β refers to the proportion of spatial resolution of P and H_{lr}. The upsampled HSI $H_{\text{up}} \in R^{\beta M \times \beta N \times B}$ has the same size as PANI. The desired HR-HSI is denoted as $H_{\text{hr}} \in R^{\beta M \times \beta N \times B}$, and the corresponding reference image is $\hat{H}_{\text{hr}}$.

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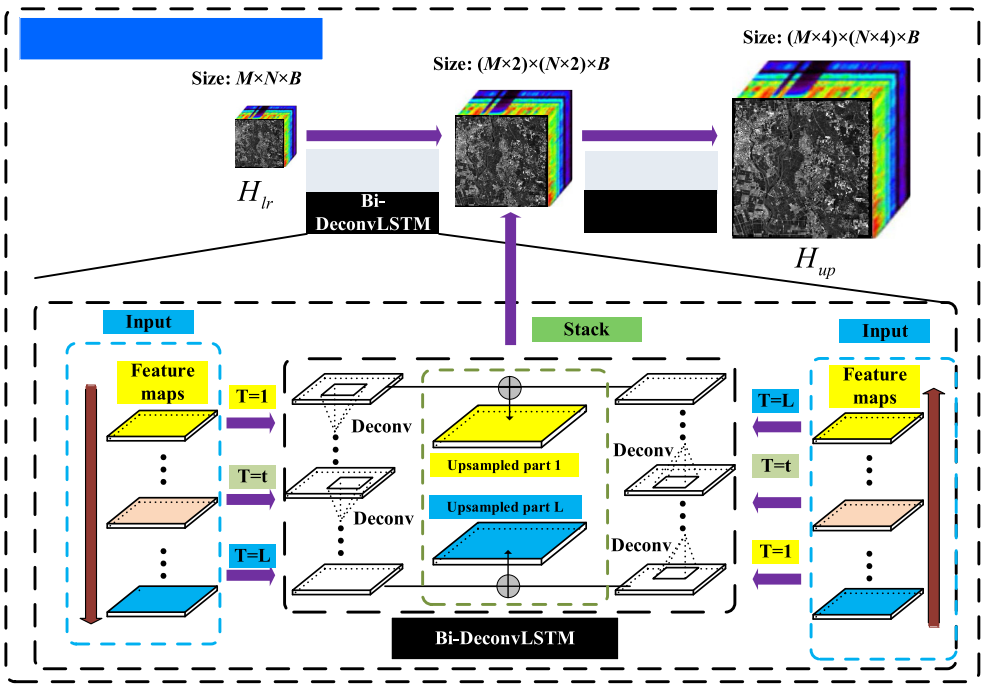


Fig. 2. Detailed architecture of DLSTM.

B. Bidirectional and Long-Term Learning-Based Upsampling

LR-HSI upsampling is fundamental to HP, where LR-HSI is aligned to the same spatial size of PANI for better spatial–spectral fusion. In HP, there are two common ways: one is using interpolation methods, such as bilinear or bicubic, and the other introduces CNN-based frameworks from the multispectral pansharpening domain [7], [13]. General interpolations are applied to LR-HSI in a band-by-band way, and then these processed bands are stacked to obtain the upsampled LR-HSI, which often brings non-negligible spectral distortion. The CNN-based upsampling methods usually use convolutional or deconvolutional layers with fixed sizes of kernels. Actually, HSIs are densely sampled from the visible to infrared spectrum, and hundreds of spectral bands are involved in processing. Thus, long-range spectral dependencies are vital for HSI representation. Limited by local receptive field, this kind of method fails to use non-local long-term spectral information, which may lead to insufficient spectral representation and spectral distortion.

In the HSIs, there is high correlation among adjacent spectral bands. Furthermore, some spectral bands distanced from each other also have non-local and long-term correlations. LSTMs were proposed to extract semantic features of long periods for the text via several gate functions. Based on LSTMs, convolutional LSTMs (ConvLSTMs) were developed by replacing the full connection of LSTMs with the convolution operation, which adapts to extract spatiotemporal features from videos. To make full use of complementary spatial–spectral information from future and previous recursion, new Bi-deconvolutional LSTMs (deconvLSTMs) are designed for LR-HSI upsampling. Bi-deconvLSTMs merge the deconvolutional process into a bidirectional structure, which extracts long-term bidirectional spatial–spectral features for upsampling.

As shown in Fig. 2, the LR-HSI is first processed by successive convolutional layers. Then, the convolutional features of LR-HSI are copied and reversed. The original and the reversed features are divided into several subsequences. These features are fed into two deconvLSTMs, respectively, which learn bidirectional, long-range spatial–spectral features for fusion. Finally, the outputs of two deconvLSTMs are stacked for the next process. This architecture is implemented repeatedly to achieve multifold resolution elevation.

Specifically, the LR-HSI H_{lr} is used as the input. First, three convolutional layers are used to extract features $F(H_{\text{lr}})$ from the input. Then, these features are divided into L subsequences in positive order along the spectral dimension, which are denoted as $(F(H_{\text{lr}})^1, \ldots, F(H_{\text{lr}})^t, \ldots, F(H_{\text{lr}})^L)$. Then, the subsequences are fed into successive recurrent cells of deconvLSTMs. Take the tth moment as an example, current input $F(H_{\text{lr}})^t$, last memory cell c_{t-1}, and the hidden state h_{t-1} are processed by the deconvolutional operation. After deconvolution, the outputs are fed to the memory cell c_t. The memory cell selectively gathers or discards the spatial–spectral information via the input gate i_t and forget gate f_t. As a result, the outputs $O_1(F(H_{\text{lr}})^1), \ldots, O_t(F(H_{\text{lr}})^t), \ldots, O_L(F(H_{\text{lr}})^L)$ of each time step are used as a reconstructed part of LR-HSI. These outputs only contain a part of information of extracted features in a specific order before the tth moment, which can be seen as pass information. From the perspective of the overall data structure, the upcoming information after the tth moment is the supplement to the information before the tth moment. Thus, it is logical to construct the other deconvLSTMs to learn the reverse input sequence to fully exploit the complementary information. Thus, the features $F(H_{\text{lr}})$ are reversed and divided as $(F(H_{\text{lr}})_r^L, \ldots, F(H_{\text{lr}})_r^t, \ldots, F(H_{\text{lr}})_r^1)$, which is processed by DeconvLSTMs in the same

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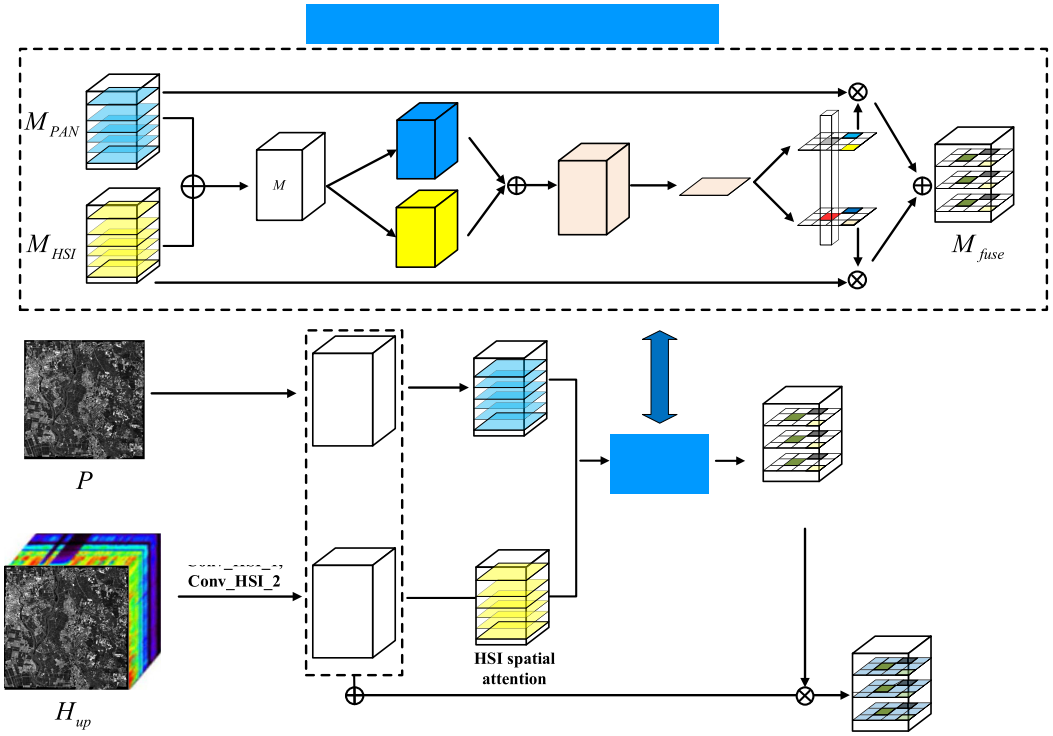


Fig. 3. Detailed architecture of the spatial branch with AIAM.

way. The corresponding outputs are denoted as $O'_1(F(H_{\text{lr}})^L_r), \ldots, O'_t(F(H_{\text{lr}})^t_r), \ldots, O'_L(F(H_{\text{lr}})^1_r)$. Since two DeconvLSTMs learn the same data in different perspectives, the corresponding outputs of every moment are element-wise summed and then stacked to generate upsampled LR-HSI H_{up}. It is formulated as

$$H_{\text{up}} = S\big(O_1\big(F(H_{\text{lr}})^1\big) + O'_1\big(F(H_{\text{lr}})^L_r\big), \ldots, \ldots, O_t(F(H_{\text{lr}})^t) + O'_t\big(F(H_{\text{lr}})^t_r\big), \ldots, O_L(F(H_{\text{lr}})^L) + O'_L\big(F(H_{\text{lr}})^1_r\big)\big) \tag{1}$$

where $S(\cdot)$ is a stack operation.

Depending on the scaling demand, H_{up} is upsampled repeatedly by the Bi-deconvLSTMs to further improve the image resolution. Finally, the final upsampled LR-HSI H_{up} with the same scale as PANI is used for fusion. DeconvLSTMs not only leverage deconvolution operation to upsample spatial–spectral information for reconstruction but also model non-local bidirectional spectral dependencies by selectively enhancing or discarding the information.

C. Divide-and-Conquer Based Spatial–Spectral Fusion

After acquiring the upsampled LR-HSI, PANI and upsampled LR-HSI are fused in this section. The HSI is a data structure that combines spectral and spatial information. The spatial dimension reflects the geometric properties, such as boundary, texture, and spatial relationships of ground objects. The spectral dimension corresponds to the radiation properties of ground objects. The physical properties of these two dimensions are quite different. Thus, it is necessary to divide the fusion along spatial and spectral dimensions and conquer the reconstruction of different dimensions with specific network architectures.

Considering the discussion above, a new DCN is proposed. It consists of two branches, spatial branch and spectral branch. The reconstruction of spatial information is realized by a novel AIAM. Since the resolution of PANI and LR-HSI is different, these two kinds of images can provide complementary multiscale spatial information. To emphasize more discriminative spatial details, the spatial attention maps of PANI and upsampled LR-HSI are generated. Then, in AIAM, these attention maps are fused to selectively focus on spatial information with different scales from PANI and upsampled LR-HSI. As for spectral reconstruction, three successive 1-D convolutional layers are stacked to fuse the spectral information among PANI and upsampled LR-HSI. The outputs of these two branches are fused to generate HR-HSI.

The architecture of the spatial branch in DCN is shown in Fig. 3. The spatial branch takes P and H_{up} as inputs. Then, two convolution layers are applied to them, and the features of P and H_{up} are obtained and denoted as $F_p(P)$ and $F_{\text{hsi}}(H_{\text{up}})$, respectively. The spatial attention map $M_{\text{up-HSI}}$ of H_{up} and the spatial attention map M_{PAN} of P are generated from $F_p(P)$ and $F_{\text{hsi}}(H_{\text{up}})$, respectively, via convolution operation with a softmax activation function.

After that, $M_{\text{up-HSI}}$ and M_{PAN} are fused in AIAM according to selective attention maps a and b. Specifically, $M_{\text{up-HSI}}$ and M_{PAN} are first combined via an element-wise summation

$$M_{\text{add}} = M_{\text{up-HSI}} + M_{\text{PAN}}. \tag{2}$$

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Then, two groups of features are obtained via different convolution operations on M_{add}, and these two groups of features are added $U = F_1(M_{\text{add}}) + F_2(M_{\text{add}})$. Afterward, the features are compacted by applying two convolutional layers on U, and the compact features s in the form of a matrix are acquired by $s = F_3(U)$.

Based on the compact matrix s, two selective attention maps a and b are generated for $M_{\text{up-HSI}}$ and M_{PAN}, respectively. Take the value at the coordinate (i, j) for example, $a(i, j)$ and $b(i, j)$ are calculated

$$a(i,j) = \frac{e^{A_j s_i}}{e^{A_j s_i} + e^{B_j s_i}}, \quad b(i,j) = \frac{e^{B_j s_i}}{e^{B_j s_i} + e^{A_j s_i}} \tag{3}$$

$$a(i,j) + b(i,j) = 1 \tag{4}$$

where $A, B \in R^{\beta M \times \beta N}$ are the learnable matrixes, A_j is the jth column of A, and s_i is the ith row of s. Finally, the fused attention map is obtained through weighted summation of spatial attention maps according to selective attention maps

$$M_{\text{fuse}} = a \cdot M_{\text{PAN}} + b \cdot M_{\text{up-HSI}}. \tag{5}$$

In the final stage, $F_p(P)$ and $F_{\text{hsi}}(H_{\text{up}})$ are multiplied by the fused spatial attention map to achieve spatial reconstruction. The final output of the spatial branch is represented as

$$H_{\text{spa}} = (F_p(P) + F_{\text{hsi}}(H_{\text{up}})) \cdot M_{\text{fuse}}. \tag{6}$$

For the spectral branch in DCN, three 1-D convolution layers are used to reconstruct the spectral information H_{spec} from PANI and upsampled LR-HSI. Finally, the spatial reconstruction H_{spa} and spectral reconstruction H_{spec} are combined

$$H_{\text{lr}} = H_{\text{spa}} + H_{\text{spec}}. \tag{7}$$

DCN-based spatial–spectral fusion takes different characteristics of spatial and spectral reconstructions into consideration, which fits the data structure of the HR-HSI well and provides the potential for performance improvement

D. Epsilon-Constraint-Based Multiobjective Optimization

Most deep-learning-based HP methods usually adopt a single objective of independent terms or multiple mixed terms. The HP task needs to optimize two objectives of both spatial and spectral preservation. These two objectives are often conflict, incomparable, and non-commensurable. Thus, it is difficult to achieve a balance between these two objectives.

Fig. 4 illustrates an example of the optimization of two conflicting objectives for the network. The path trajectory in black is the procedure of searching for the optimal solution. When the network is optimized by a single objective individually, it is easy for the network to converge to a desired set of optimization solutions. Two red nodes represent the corresponding optimal solutions under different single objectives. However, when two objectives are involved, these two objectives will combat and cause the oscillation of performance. The network is prone to train the easy-to-optimize objective and achieve undesirable performance on the other. Hence, to meet both the objectives, tradeoff Pareto-optimal solutions, marked in yellow node, should be considered.

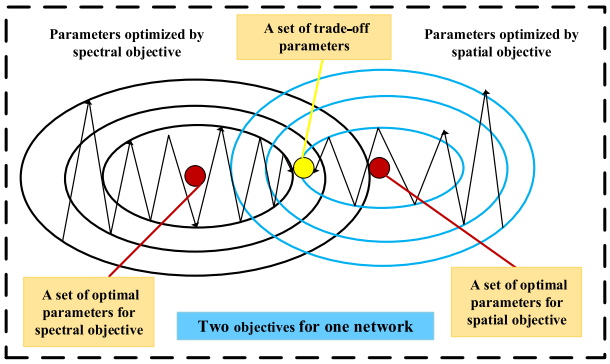


Fig. 4. Example of optimization of two conflicting objectives for the network.

Concerning the above issue, the pansharpening problem is formulated as an Epsilon-constraint-based multiobjective optimization. In the Epsilon constraint method, two objective functions related to the spatial and spectral evaluation are defined. SAM is selected as the first objective function $f_1(H_{\text{hr}}, \hat{H}_{\text{hr}})$. It is calculated by (8). The smaller SAM value indicates that the network maintains better performance in spectral preservation

$$f_1(H_{\text{hr}}, \hat{H}_{\text{hr}}) = \text{SAM}(H_{\text{hr}}, \hat{H}_{\text{hr}}) = \text{across}\left(\frac{\langle H_{\text{hr}}, \hat{H}_{\text{hr}}\rangle}{\|H_{\text{hr}}\|_2 \left\|\hat{H}_{\text{hr}}\right\|_2}\right). \tag{8}$$

By minimizing the f_1 loss function, the network would have extraordinary preservation in the spectral domain, and a set of optimal parameters θ_1 for the spectral objective are obtained.

After training in the spectral domain, the network is expected to have a good spatial preservation ability simultaneously. The second objective function is designed using CC. The larger the value of CC, the less spatial distortion the synthetic HR-HSI has. Here, the inverse of CC is defined as the second objective function

$$f_2(H_{\text{hr}}, \hat{H}_{\text{hr}}) = \frac{1}{\text{CC}} = \frac{\sqrt{\sum_{h,w}[(H_{\text{hr}} - \mu)^2] \sum_{h,w}[(\hat{H}_{\text{hr}} - \hat{\mu})^2]}}{\sum_{h,w}[(H_{\text{hr}} - \mu)(\hat{H}_{\text{hr}} - \hat{\mu})]} \tag{9}$$

where μ and $\hat{\mu}$ are the mean of the fused HR-HSI H_{hr} and the reference HR-HSI $\hat{H}_{\text{hr}}$, respectively.

Actually, it is not appropriate to use f_2 to optimize the network directly. When an extra conflicting objective function is applied, the ability of spectral information preserving of the network must be degraded, which is not desirable. Hence, to ensure the reduction of spectral preservation ability as less as possible, a spectral constraint is added to the second objective. The second objective function is redefined as follows:

$$f_2 = \frac{1}{\text{CC}} + \max\{0, \text{SAM}(H_{\text{hr}}, \hat{H}_{\text{hr}}) - \varepsilon\}. \tag{10}$$

The spectral constraint measures how far the current SAM is from the ideal SAM. The aim of f_2 is to improve the spatial preservation ability and reduce the spectral reconstruction as

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TABLE I
PROCEDURE OF THE PROPOSED MO-DCN METHOD

INPUT Low resolution hyperspectral image H_{lr}, panchromatic image P
Output High resolution hyperspectral image H_{hr}
//network inference
Begin
input H_{lr} to DLSTM to obtain the upsampled hyperspectral image H_{up}
input the H_{up} and P to spatial branch of DCN to obtain H_{spa}
input the H_{up} and P to spectral branch of DCN to obtain H_{spec}
add H_{up} and H_{spec} up
END
OUTPUT: the fused HR-HSI H_{hr} is obtained
// optimization
Begin
randomly initialize weight matrices and biases of MO-DCN
for Q **epochs**
  for E epochs
    compute the loss function by (8)
    update the parameters by minimizing the loss (8)
  calculate ε by (11)
  end for
  for F epochs
    compute the loss function by (10)
    update the parameters by minimizing the loss (10)
  end for
end for
END

little as possible. In (10), when the value of the current SAM diverges from the ideal SAM greatly, a higher penalty value is added to urge the network to focus more on spectral preservation. The ideal SAM is denoted as ε, which is the lower bound of spectral preservation. It is defined as the SAM value by only optimizing f_1 function

$$\varepsilon = \mathrm{SAM}(H_{\mathrm{hr}}, \hat{H}_{\mathrm{hr}}|\theta_1) = \mathrm{across}\left(\frac{\langle H_{\mathrm{hr}}, \hat{H}_{\mathrm{hr}}\rangle}{\|H_{\mathrm{hr}}\|_2 \left\|\hat{H}_{\mathrm{hr}}\right\|_2}\right) \quad (11)$$

where $\mathrm{SAM}(H_{\mathrm{hr}}, \hat{H}_{\mathrm{hr}}|\theta_1)$ denotes the SAM value of the network under the parameters θ_1.

The whole network is trained alternately by f_1 and f_2 loss functions. The detailed procedure of MO-DCN is listed in Table I. MO-DCN combines bidirectional and long-term learning-based upsampling and divide-and-conquer-based spatial–spectral fusion into a unified network. By applying the Epsilon-constraint-based multiobjective optimization for the whole network, such a pattern of mutual restraint can effectively alleviate the conflict between the goals and achieve an excellent compromise between spectral and spatial preservation for HP.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, the effectiveness of the proposed method is validated on four benchmark HSIs. The fusion performance of the proposed method is validated qualitatively and quantitatively by comparison with ten representative methods. The experiments are divided into six parts, data description, quality assessment, experimental setups, experimental results on simulated dataset, experimental results on real dataset, and effectiveness analysis of each step.

A. Data Description

Four HSI datasets including the Botswana dataset, the Chikusei dataset, the Pavia center dataset, and the Los Angeles dataset are used, which are captured by different sensors. The original HSI is regarded as the ground truth, and the training dataset and testing dataset are generated according to Wald's protocol [15]. Specifically, the LR-HSI is obtained using a Gaussian kernel and downsampling on the original HSI. The PANI is simulated by averaging the visible bands in the approximate range of 400–760 nm. Then, paired LR-HSI and PANI are divided into several cubic patches for training and testing. For all the datasets, the size of Gaussian kernels is 7×7. For the Botswana dataset, the Chikusei dataset, and the Pavia center dataset, the factor of downsampling is 4, while the factor of downsampling is 3 for the Los Angeles dataset. More detailed information of three datasets is listed as follows:

1) Botswana Dataset: The Botswana dataset was taken over the Okavango Delta, Botswana, by the Hyperion sensor on the EO-1 satellite. The image has 1496×256 pixels with 242 bands covering 400–2500 nm. There are a total of 145 bands after discarding the uncalibrated and noisy bands. In our experiments, a sub-image from this image with a spatial coverage of 1280×256 is used. After processing, the degraded HSI is obtained with $320 \times 64 \times 145$ pixels, and the corresponding PANI has 1200×240 pixels with a spatial resolution of 7.5 m. The degraded HSI is divided into 20 cubic patches of size $40 \times 40 \times 145$ without overlap. The training set comprises 12 patches. The size of LR-HSI patches is $40 \times 40 \times 145$, and the corresponding size of PANI patches is 120×120. The rest eight LR-HSI patches and PANI patches constitute the testing set.

2) Chikusei Dataset: The Chikusei dataset is an aviation scene of Chikusei, Ibaraki, Japan, which was captured by Headwall's Hyperspec Visible and Near-Infrared, series C (VNIR-C) imaging sensor on July 29, 2014. The scene consists of 2517×2235 pixels, and the ground sampling distance is 2.5 m. It is made up of 128 bands covering a spectral range from 363 to 1018 nm. Only a region with 1280×1280 spatial pixels from the original scene is used. The degraded HSI contains 320×320 spatial pixels with a spatial resolution of 10 m, and the corresponding PANI has four times higher spatial resolution of 2.5 m with 1280×1280 spatial pixels. The training set and testing set are obtained by dividing the degraded HSI into 25 cubic patches of size $64 \times 64 \times 128$ without overlap. Accordingly, the PANI is divided into the same number of patches with a size 256×256. About 15 cubic patches (about 60%) are selected randomly to compose the training set, and the rest is used for testing.

3) Pavia Center Dataset: This dataset presents a scene of the city of Pavia, Italy, which was collected by the reflective optics system imaging spectrometer (ROSIS). Due to the

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TABLE II
QUANTITATIVE INDICES OF FUSION PERFORMANCE

Type	Indices	Statement	Optimal value	Definition
Spatial quality	CC	Measurement of spatial distortion	1	$\mathrm{CC}=\frac{\sum_{\beta M,\beta N}[(H_{hr}-\mu)(\hat{H}_{\beta M,\beta N}-\hat{\mu})]}{\sqrt{\sum_{\beta M,\beta N}[(H_{hr}-\mu)^2\sum_{\beta M,\beta N}[(\hat{H}_{hr}-\hat{\mu})^2]}}$
	D_s	Measurement of spatial distortion when reference image is not available	0	$D_s=\sqrt[\varepsilon]{\frac{1}{b}\sum_{q=1}^{b}\left\|Q(H_{lr}^{q},P)-Q(H_{lr}^{q_1},P_{lr})\right\|^{\varepsilon}}$
Spectral quality	SAM	Measurement of spectral distortion	0	$\mathrm{SAM}=across\left(\frac{\langle H_{hr},\hat{H}_{hr}\rangle}{\|H_{hr}\|_2\|\hat{H}_{hr}\|_2}\right)$
	D_λ	Measurement of spectral distortion when reference image is not available	0	$D_\lambda=\sqrt[\varepsilon]{\frac{1}{b(b-1)}\sum_{q_1=1}^{b}\sum_{q_2=1,q_2=q_1}^{b}\left\|Q(H_{lr}^{q_1},H_{lr}^{q_2})-Q(H_{lr}^{q_1},H_{lr}^{q_2})\right\|^{\varepsilon}}$ $Q(I,J)=\frac{\varphi(I,J)}{\varphi(I)\varphi(J)}\frac{2\bar{I}\bar{J}}{\bar{I}^2+\bar{J}^2}\frac{2\varphi(I)\varphi(J)}{\varphi(I)^2+\varphi(J)^2}$
Comprehensive quality	RMSE	How close the fused image to the reference one	0	$\mathrm{RMSE}=\sqrt{\frac{1}{\beta M\beta N}\sum_{\beta M,\beta N}\left(H_{hr}-\hat{H}_{hr}\right)^2}$
	ERGAS	Global quality evaluation of the fused image	0	$\mathrm{ERGAS}=100\beta\sqrt{\frac{1}{b}\sum_{q=1}^{b}\left[\frac{RMSE_q}{\mu(H_{hr}^{q})}\right]^2}$
	PSNR	Assess the fusion quality of each band	∞	$\mathrm{PSNR}=\frac{1}{B}\sum_{q=1}^{B}\left[10\cdot\log_{10}\left(\frac{\max(H_{hr}^{q})}{RMSE_q}\right)^2\right]$
	QNR	A comprehensive evaluation of spatial and spectral quality when reference image is not available	1	$QNR=\left(1-D_\lambda\right)^{\eta}\left(1-D_s\right)^{\rho}$

removal of noisy bands and the discarding of poor-quality areas, the original image with a size of $1096\times1096\times115$ is cropped to $960\times640\times102$. The degraded HSI is generated with $240\times160\times102$ pixels, and the corresponding PANI has 960×640 pixels with a spatial resolution of 1.3 m. The degraded image is partitioned into 24 cubic patches with the size $40\times40\times102$, and the corresponding size of PANI patches is 160×160. We randomly select 14 patches (about 60%) for training and the remaining ten for testing.

4) Los Angeles Dataset: This dataset describes a port of Los Angeles. Two instruments of the EO-1 satellite, the Hyperion and the Advanced Land Imager (ALI), are used to obtain a real LR-HSI with a corresponding PANI. The LR-HSI with a size of $120\times120\times145$ has 242 bands with a spatial resolution of 30 m, and the PANI with a size of 360×360 has a spatial resolution of 10 m. After discarding uncalibrated and noisy bands, 145 bands are left. The dataset is divided into nine patches with the size $40\times40\times145$ and the corresponding size of PANI patches is 120×120.

B. Quality Assessments

The performances of different methods are evaluated qualitatively and quantitively. Qualitative assessment is realized by the visualization of pansharpening HSI of test samples. Quantitative results are evaluated by eight widely used indices, including CC, SAM, erreur relative globale adimensionnelle de synthèse (ERGAS), peak SNR (PSNR), D_λ, D_s, and quality with no reference (QNR). Table II provides the brief introduction in a tabular form. These different quantitative measurements evaluate the fusion results from spatial, spectral, and global preservation performance, where CC is for spatial evaluation, SAM and PSNR for spectral evaluation, and RMSE and ERGAS for a comprehensive evaluation. Besides, since there is no reference image available for the real dataset, three indices are used to measure the quality of fused image in the generalization experiment. In Table II, $\hat{H}_{\mathrm{hr}}^{q}$ is the qth-band of $\hat{H}_{\mathrm{hr}}$. ε and κ are typically set to be 1. In $Q(I, J)$, $\varphi(I, J)$ is designed to calculate the covariance between I and J, $\varphi(I)$ is the variance of I, and $\bar{I}$ is the sample mean. η and ρ in the QNR index are two coefficients.

C. Experimental Setups

To validate the effectiveness of the MO-DCN method, ten state-of-the-art or representative HSI pansharpening methods participate in the comparison: IHS [21], SFIM [26], GS [20], MTF-GLP [25], MTF-GLP-HPM [23], guided filter principal

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TABLE III
DETAILED STRUCTURE OF MO-DCN

Part	No	Name	Type of Layer	Kernel Size	Activation
DLSTM	1	Conv1	Convolutional	$3\times3\times B$	ReLu
	Belong to layer 1	Forward	DeconvLSTM	$3\times3\times B$	Tanh/Sigmoid
	Belong to layer 1	Backward	DeconvLSTM	$3\times3\times B$	Tanh/Sigmoid
	2	Conv2	Convolutional	$3\times3\times B$	ReLu
	Belong to layer 2	Forward	DeconvLSTM	$3\times3\times B$	Tanh/Sigmoid
	Belong to layer 2	Backward	DeconvLSTM	$3\times3\times B$	Tanh/Sigmoid
Spatial branch	1	Conv_PAN_1	Convolutional	$3\times3\times64$	ReLu
	2	Conv_PAN_2	Convolutional	$3\times3\times B$	ReLu
	3	Conv_HSI_1	Convolutional	$3\times3\times64$	ReLu
	4	Conv_HSI_2	Convolutional	$3\times3\times B$	ReLu
	5	Conv_3	Convolutional	$3\times3\times64$	ReLu
	6	Conv_4	Convolutional	$3\times3\times128$	ReLu
	7	Conv_5	Convolutional	$3\times3\times256$	ReLu
	8	Conv_6	Convolutional	$3\times3\times256$	ReLu
	9	Conv_7	Convolutional	$3\times3\times128$	ReLu
	10	Conv_8	Convolutional	$3\times3\times1$	ReLu
Spectral branch	1	Conv1	Convolutional	$1\times1\times B$	ReLu
	2	Conv2	Convolutional	$1\times1\times B$	ReLu

component analysis (GFPCA) [36], GSA [19], CNMF [29], PANnet [37], and HP framework based on a 3-DGAN (HPGAN) [17]. About 20% of the training samples are used as the validation set for parameter tuning. In SFIM, the kernel size and the initial values of the kernel are determined using the trial-and-error procedure in the validation set. As suggested in [24], the Gaussian kernels of MTF-GLP and MTF-GLP-HPM are designed with a size of 9×9. In GFPCA, the sizes of guided filters are set to the default value [35]. As for CNMF, the threshold of change ratio in the inner loop and the outer loop for LR-HSI unmixing is set to be 1×10^{-8} and 1×10^{-2}, respectively. For PANnet, HPGAN, and MO-DCN, the hyper-parameters, such as batch size, learning rate, and epoch number, are determined using trial-and-error procedures. In MO-DCN, mini-batch gradient descent is adopted with the size of mini-batch 8, learning rate 0.002, and epochs 400. The detailed design of MO-DCN is listed in Table III. It records the structure of DLSTM, spatial, and spectral branch of DCN. All the experiments were carried out using Python language and Pytorch library on an NVIDIA 2080Ti graphics card.

D. Experimental Results on Simulated Datasets

In this section, the qualitative and quantitative results of all the methods on three datasets are presented. The quantitative results are evaluated in CC, SAM, RMSE, ERGAS, and PSNR. The best results of each index are highlighted in bold. For the qualitative results, one representative fused result from the testing set of each dataset is selected to make a visual comparison. To better observe the detailed reconstruction, a representative region with rich texture in a yellow square is enlarged.

TABLE IV
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT HP METHODS ON THE CHIKUSEI DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
IHS	0.8432	6.1794	0.0402	12.5887	27.9833
SFIM	0.8790	4.8276	0.0315	8.0896	30.1625
GS	0.7575	6.0853	0.0469	8.4277	26.6858
MTF-GLP	0.8637	5.5994	0.0393	6.8517	28.1811
MTF-GLP-HPM	0.8527	4.9522	0.0379	10.9081	28.4628
GFPCA	0.8616	5.5725	0.0369	7.1574	28.7118
GSA	0.8643	4.6547	0.0327	8.1159	29.7430
CNMF	0.9098	4.4717	0.0301	6.9039	30.5363
PANnet	0.9108	4.4793	0.0287	5.7158	31.0080
HPGAN	0.9247	4.1631	0.0269	4.8725	31.4094
MO-DCN	**0.9417**	**3.4151**	**0.0234**	**4.3033**	**32.9229**

1) Chikusei Dataset: The detailed quantitative results of 11 fusion methods on the Chikusei dataset are recorded in Table IV. As shown in IV, CNMF obtains higher CC values and performs better in spatial preservation than other traditional methods, IHS, SFIM, GS, MTF-GLP, MTF-GLP-GPM, GFPCA, and GSA. Among all the traditional methods, GSA achieves the least spectral distortion in terms of SAM. Compared with the traditional methods, PANnet and HPGAN

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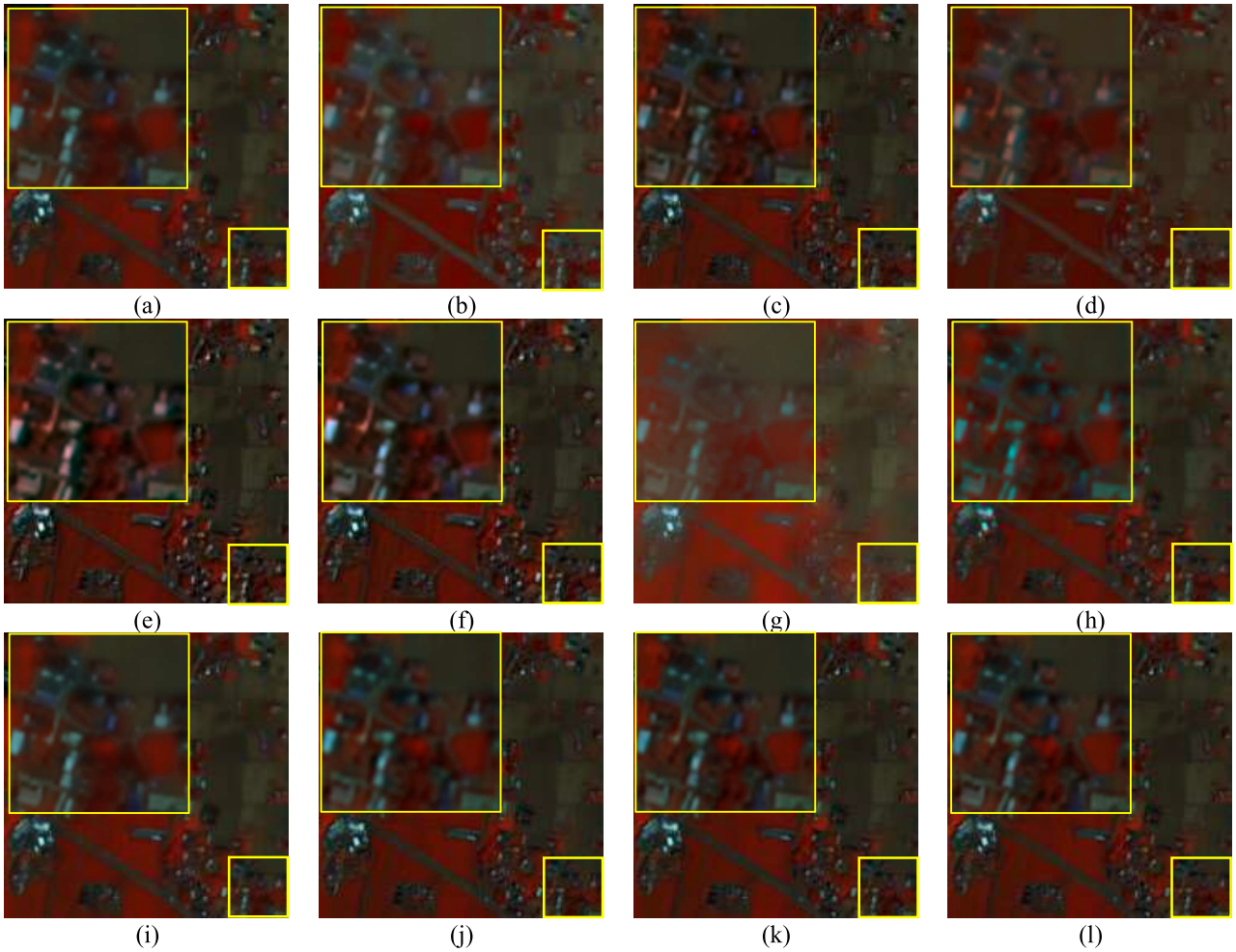


Fig. 5. Ground truth and visualization of a representative testing sample on the Chikusei dataset. (a) Ground truth. (b) IHS. (c) SFIM. (d) GS. (e) MTF-GLP. (f) MTF-GLP-HPM. (g) GFPCA. (h) GSA. (i) CNMF. (j) PANnet. (k) HPGAN. (l) MO-DCN.

are more outstanding in reconstructing spatial information via a series of deep feature extraction. However, these two methods are not as good as MO-DCN in terms of SAM. MO-DCN achieves much less spectral distortion with 0.748 improvement in SAM through modeling long-term spectral dependencies. Compared with PANnet and HPGAN, MO-DCN has significant improvement in spatial enhancement via selective attention fusion of AIAM. What is more, MO-DCN has obvious advantages in terms of comprehensive indices, RMSE and ERGAS. The visualization of fusion results of all the methods is shown in Fig. 5. The left-top patch is the enlarged view of the corresponding small patch in the right bottom. For MTF-GLP-HPM, as shown in Fig. 5(f), the color of the buildings is much different from that of ground truth and the image is over-pansharpened. In Fig. 5(c), there is an obvious blue noisy point in the enlarged area. IHS, GS, and GFPCA fail to restore many details and cause non-negligible spatial blurring, as shown in Fig. 5(c), (d), and (g). Among all the methods, the result of the proposed MO-DCN method is the most similar to the ground truth. MO-DCN obtains better fusion results through powerful boundary preservation ability.

2) Botswana Dataset: For the Botswana dataset, the quantitative fusion results are summarized in Table V. IHS, MTF-GLP, and CNMF maintain high CC values but have higher SAM values. It reveals that these methods fail to preserve spatial and spectral information simultaneously.

TABLE V
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT HP METHODS ON THE BOTSWANA DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
IHS	0.8567	9.5507	0.0374	5.1285	28.6181
SFIM	0.9160	3.7161	0.0408	3.7560	27.8727
GS	0.9339	3.1123	0.0341	3.6314	29.4094
MTF-GLP	0.9127	4.2563	0.0458	5.3643	26.9662
MTF-GLP-HPM	0.8841	3.1709	0.0315	2.8986	30.3201
GFPCA	0.8875	3.0201	0.0335	3.0868	29.7095
GSA	0.9388	3.0735	0.0346	3.8731	29.2978
CNMF	0.9035	4.5502	0.0480	3.8304	26.6147
PANnet	0.9269	2.9831	0.0250	2.3138	32.2389
HPGAN	0.9333	2.8801	0.0241	2.2111	32.5318
MO-DCN	**0.9535**	**2.6815**	**0.0235**	**2.0931**	**33.7509**

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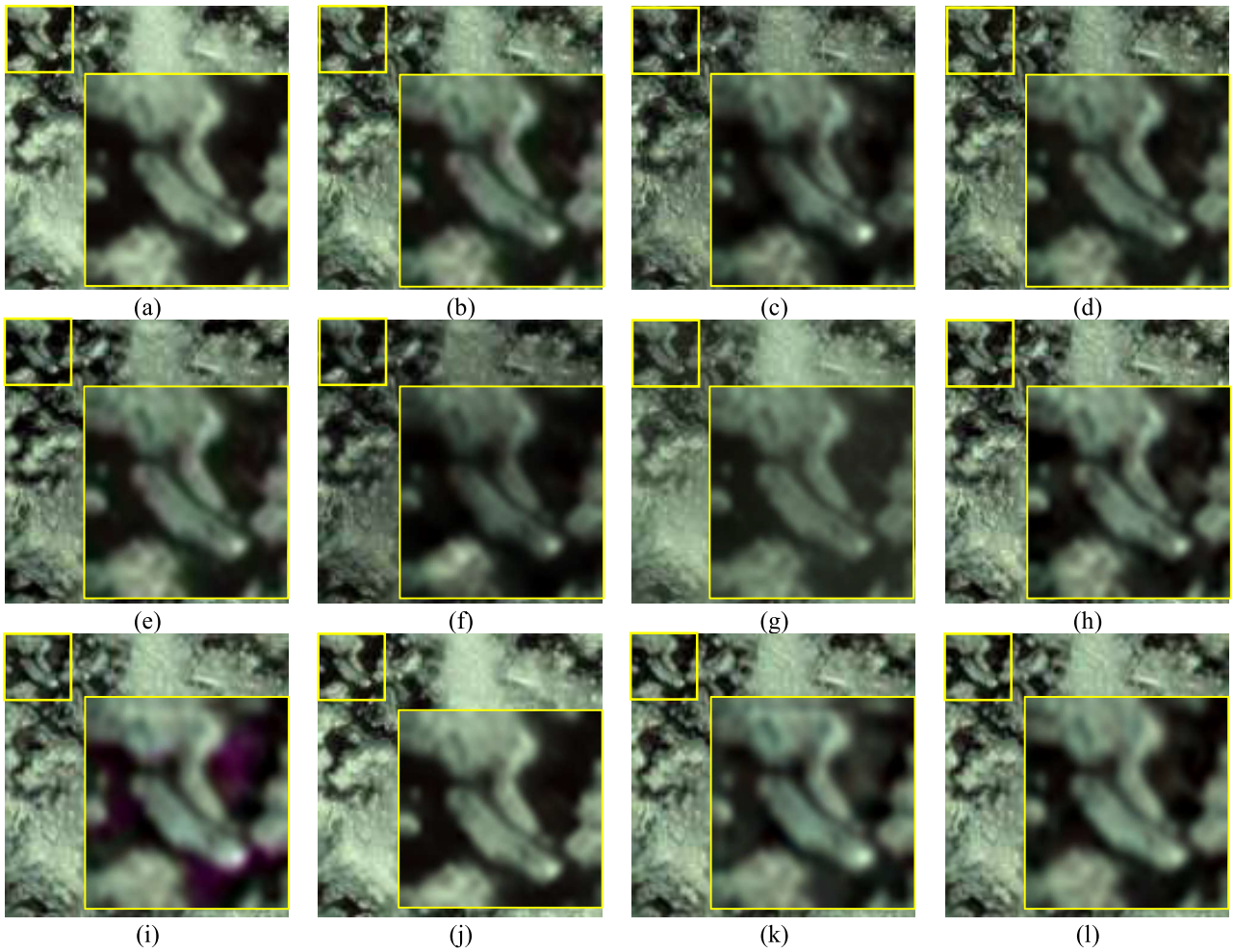


Fig. 6. Ground truth and visualization of a representative testing sample on the Botswana dataset. (a) Ground truth. (b) IHS. (c) SFIM. (d) GS. (e) MTF-GLP. (f) MTF-GLP-HPM. (g) GFPCA. (h) GSA. (i) CNMF. (j) PANnet. (k) HPGAN. (l) MO-DCN.

Deep-learning-based methods are superior to traditional methods in terms of comprehensive RMSE and ERGAS indices by excellent non-linear and hierarchical feature extraction ability. It is worth noting that the SAM index of deep-learning-based methods has improved significantly, indicating the excellent spectrum preservation ability of these algorithms. Among the deep-learning-based methods, the proposed method obtains the best fusion performance in the aspect of all the five evaluation indices. It achieves an excellent co-improvement in both spatial and spectral preservation by designing the divide-and-conquer structure and multiobjective optimization.

The visualization of HP results of different methods is displayed in Fig. 6. The reference HR-HSI is presented in Fig. 6(a). The small patch in yellow is enlarged to show clearly. It can be seen in Fig. 6(b), (e)–(h) that the results of IHS, GSA, GFPCA, MTF-GLP, MTF-GLP-HPM, and GFPCA are blurry in the boundaries of many areas since insufficient spatial details are extracted from PANI and upsampled HSI. For SFIM, there is an area at the bottom right of the enlarged patch that is brighter than the ground truth, which shows this area is oversharpened. Compared with other methods, MO-DCN has better spatial enhancement and edge preservation, where the color transition among heterogeneous regions is more natural and realistic with the reference of ground truth.

TABLE VI
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT HP METHODS ON THE PAVIA CENTER DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
IHS	0.9326	7.1330	0.0336	8.5876	29.9035
SFIM	0.9307	7.2126	0.0335	7.7801	29.6956
GS	0.9388	7.0353	0.0339	8.6153	29.7950
MTF-GLP	0.9270	7.5868	0.0373	9.6828	28.9847
MTF-GLP-HPM	0.9122	9.6614	0.0435	7.8669	27.4935
GFPCA	0.9150	7.9820	0.0414	8.6950	28.2821
GSA	0.9471	7.7950	0.0297	7.7816	30.9102
CNMF	0.9546	8.7799	0.0280	6.7988	31.4525
PANnet	0.9508	6.4705	0.0281	5.7367	31.4744
HPGAN	0.9582	6.0256	0.0259	5.2126	32.1967
MO-DCN	**0.9726**	**5.0858**	**0.0229**	**5.0437**	**33.4961**

3) Pavia Center Dataset: The statistical HP results on the Pavia dataset are recorded in Table VI. It is noticeable that the fusion performance of deep-learning-based methods has consistent advantages on all the three datasets, while the

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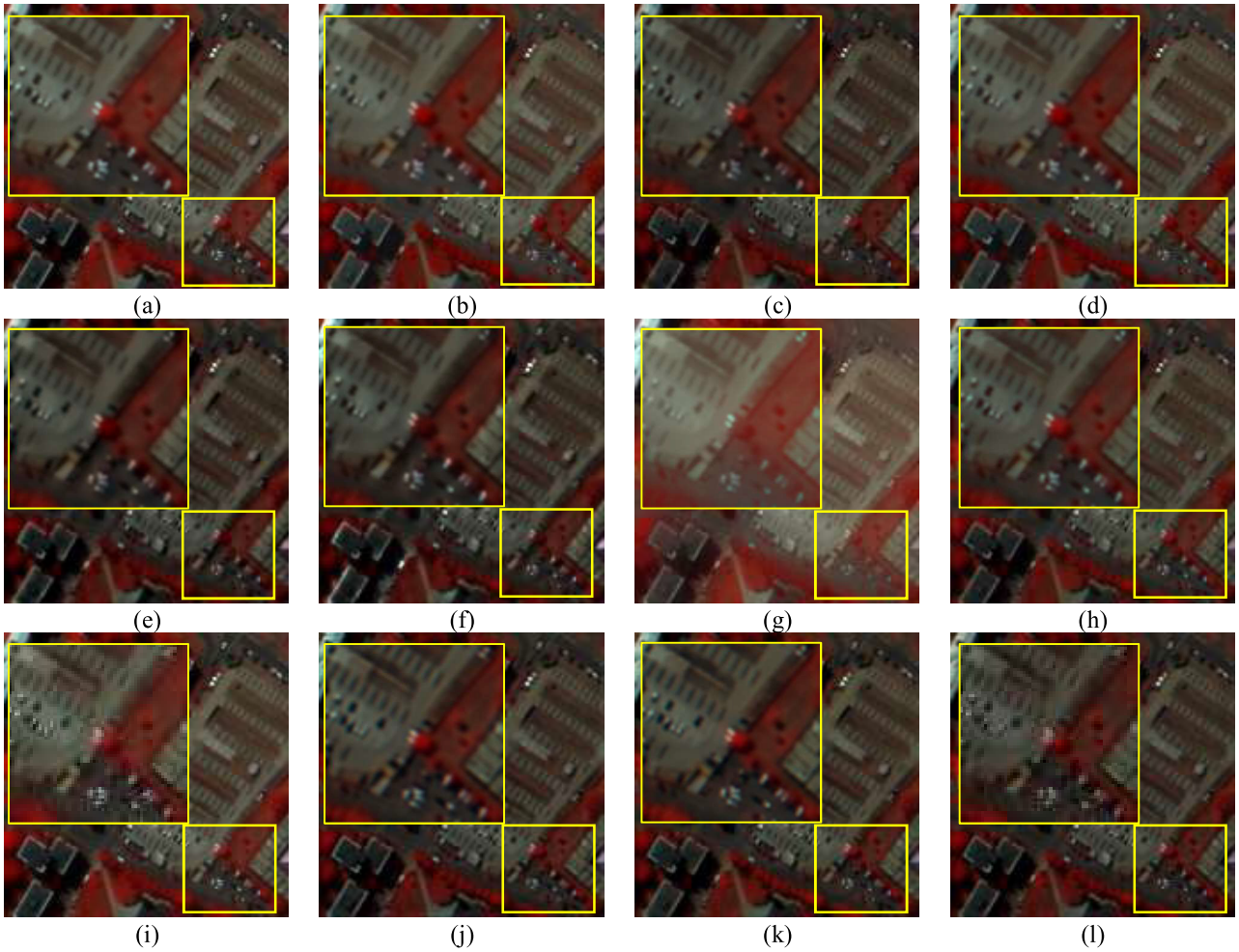


Fig. 7. Ground truth and visualization of a representative testing sample on the Pavia Center dataset. (a) Ground truth. (b) IHS. (c) SFIM. (d) GS. (e) MTF-GLP. (f) MTF-GLP-HPM. (g) GFPCA. (h) GSA. (i) CNMF. (j) PANnet. (k) HPGAN. (l) MO-DCN.

performance of most traditional methods obviously fluctuates across three datasets. For this dataset, all the methods surpass 0.90 in terms of CC indices for spatial reconstruction. However, severe spectral distortion appears in MTF-GLP-HPM, CNMF, and GFPCA. HPGAN and PANnet alleviate the spectral distortion to a certain degree by non-linear spectral information extraction. Among all the methods, MO-DCN achieves the highest fusion performance in five indices. Although MO-DCN achieves similar spatial performance to the best baseline HPGAN, MO-DCN significantly reduces the spectral distortion by the usage of effective Epsilon-constraint-based multiobjective optimization. It reveals that a better balance between spatial and spectral objectives is acquired by MO-DCN.

Fig. 7(a)–(i) visualizes the fusion results of 11 methods. By observing the vehicles in Fig. 7(d)–(f), traditional methods, such as GS, MTF-GLP, and MTF-GLP-HPM, cause zigzag boundaries for vehicles. Compared with the traditional methods, PANnet, HPGAN, and MO-DCN have excellent fine-grind spatial information extraction ability and better boundary preservation for vehicles. For MO-DCN, it is superior with boundary preservation in different scale objects, such as vehicles and buildings, and achieves the smallest distortion in spatial distortion compared with the ground truth.

TABLE VII
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE LOS ANGELES DATASET

Quantitative indices	D_λ ($\downarrow$)	D_s ($\downarrow$)	QNR ($\uparrow$)
IHS	0.1109	0.0796	0.8185
SFIM	0.0923	0.1271	0.8030
GS	0.0972	0.0743	0.8360
MTF-GLP	0.0748	0.0895	0.8499
MTF-GLP-HPM	0.0930	0.0910	0.8311
GFPCA	0.0997	0.0794	0.8330
GSA	0.1010	0.0485	0.8593
CNMF	0.0823	0.092	0.8384
PANnet	0.0866	0.0389	0.8670
HPGAN	0.1164	0.0669	0.8244
MO-DCN	**0.0637**	**0.0345**	**0.8698**

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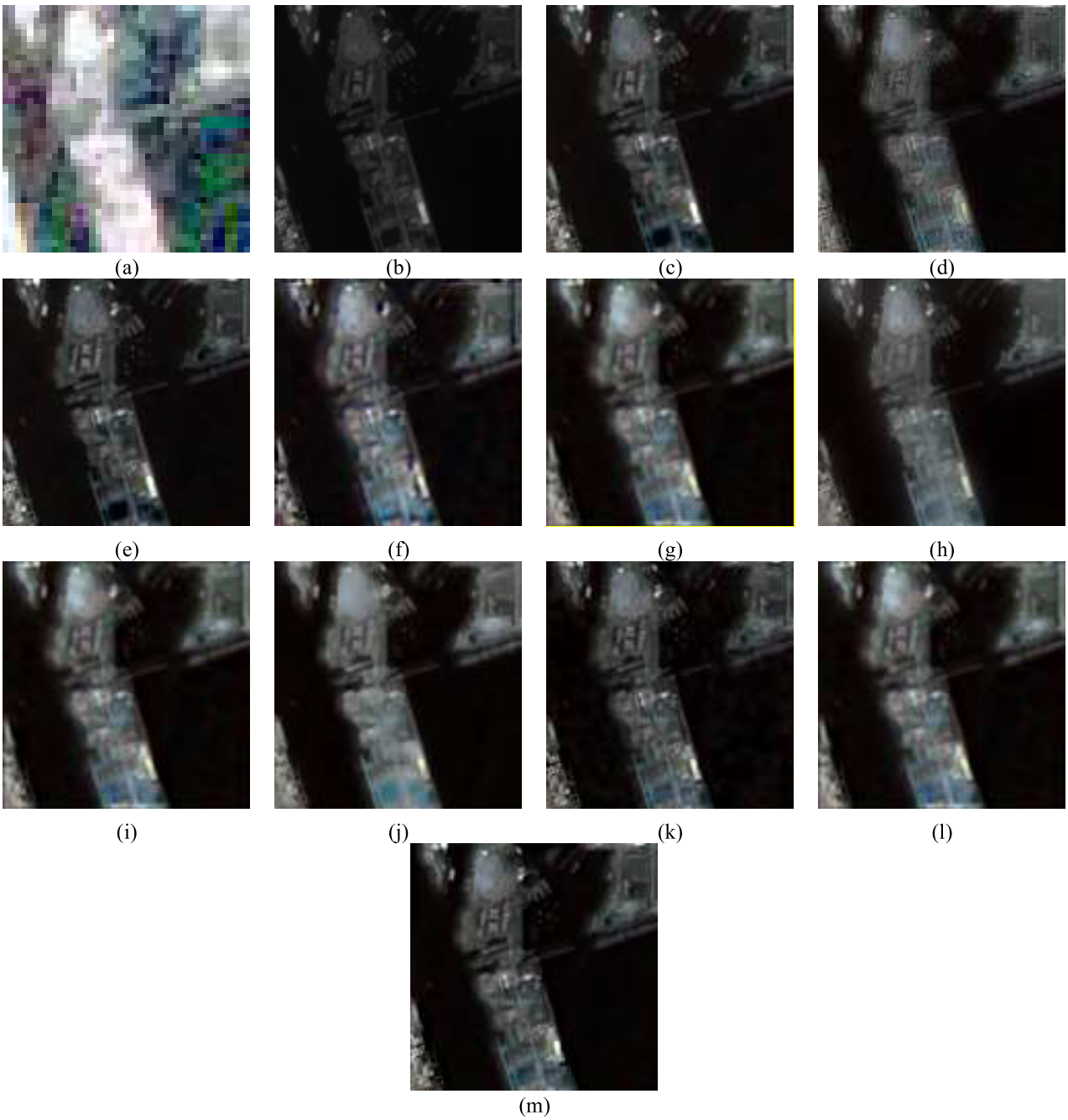


Fig. 8. Visualization of a representative testing sample on the Los Angeles dataset. (a) False color image of real HSI image. (b) Real PANI. The pansharpening results of (c) IHS, (d) SFIM, (e) GS, (f) MTF-GLP, (g) MTF-GLP-HPM, (h) GFPCA, (i) GSA, (j) CNMF, (k) PANnet, (l) HPGAN, (m) MO-DCN.

E. Experimental Results on Real Datasets

To evaluate the generation ability of the proposed method, a real hyperspectral and PANI matching dataset, the Los Angeles dataset, is used. All the comparison methods are trained on the Botswana dataset, and then the fusion performance is verified on the Los Angeles dataset. Since there are no reference HR-HSI, three no-reference metrics including D_λ, D_s, and QNR are used to provide the evaluation of fusion quality. The quantitative results of different methods are listed in Table VII. To better visualize the pansharpening results, one representative patch that contains abundant detailed information is selected, enlarged, and visualized. The results are shown in Fig. 8(a)–(m). The proposed method used recurrent neural networks (RNNs) to upsample LR-HSI, which requires fewer parameters. In addition, the dropout layer also contributes to the generalization ability of the proposed method. From the visualization results, the proposed MO-DCN method can better maintain spatial detail information than other methods. Furthermore, MO-DCN successfully enhances high-frequency components.

F. Effectiveness Analysis of Each Step in MO-DCN

There are three essential components in MO-DCN: DLSTM, DCN, and Epsilon-constraint-based multiobjective optimization. Accordingly, three groups of experiments are

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TABLE VIII
UPSAMPLING RESULTS OF DIFFERENT METHODS ON THE BOTSWANA DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
Bicubic interpolation	0.7947	2.9685	0.0410	3.7695	27.9831
LapSRN	0.8719	2.6857	0.0357	3.5857	29.1028
DLSTM	**0.8936**	**2.5495**	**0.0314**	**3.1517**	**30.1188**

TABLE IX
UPSAMPLING RESULTS OF DIFFERENT METHODS ON THE CHIKUSEI DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
Bicubic interpolation	0.8375	4.9980	0.0349	8.6991	29.2163
LapSRN	0.8643	4.6547	0.0328	8.1159	29.7430
DLSTM	**0.8991**	**3.6803**	**0.0269**	**6.7505**	**31.4962**

TABLE X
UPSAMPLING RESULTS OF DIFFERENT METHODS ON THE PAVIA CENTER DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
Bicubic interpolation	0.8209	8.9182	0.0506	10.0922	26.6167
LapSRN	0.8644	7.4077	0.4482	9.0053	27.6280
DLSTM	**0.8872**	**6.0935**	**0.0399**	**7.2408**	**29.0289**

implemented on three datasets to validate the effectiveness of three components, respectively.

1) Effectiveness Analysis of DLSTM in MO-DCN: To evaluate the effectiveness of DLSTM, a traditional bicubic interpolation and a deep-learning-based LapSRN [7] are used for comparison. The DLSTM is separated from MO-DCN and trained independently. MSE is adopted as the training objective for LapSRN and DLSTM for fair comparison.

As shown in Tables VIII–X, bicubic interpolation inserts a new pixel by gathering the information from the local spatial neighborhood, which brings high spectral distortion. LapSRN and DLSTM perform better in every index than bicubic interpolation through hierarchical spatial feature extraction and spectral modeling of CNN. Compared with LapSRN, DLSTM achieves less spectral distortion using the bidirectional structure to model long-term spectral dependencies. Meanwhile, the deconvolutional LSTM makes full use of the spatial information to better upsample LR-HSI, which obtains higher CC indices on all the three datasets.

2) Effectiveness Analysis of DCN in MO-DCN: To validate the effectiveness of DCN, single spatial branch network (SPA), spatial–spectral dual network (SPA-SPE), and spatial–spectral dual network with AIAM (SPA-SPE-AIA) are used for comparison. Actually, SPA-SPE-AIA is DCN. Tables XI–XIII show the results of different methods on three datasets. Compared with SPA, SPA-SPE significantly reduces the SAM by adding an additional spectral branch. By reconstructing spatial and spectral information in a divide-and-conquer way, SPA-SPE considers different characteristics of spatial and spectral information and elevates spectral reconstruction performance. Compared with SPA-SPE, DCN uses an AIAM to gather complementary spatial information from upsampled HSI and PANI and further promotes spatial preservation on three datasets. Among these methods, DCN behaves well in not only comprehensive indices, such as RMSE and ERGAS, but also individual indices, such as CC and SAM.

3) Effectiveness Analysis of the Epsilon-Constraint-Based Multiobjective Optimization in MO-DCN: The effectiveness of Epsilon-constraint-based multiobjective optimization is validated by comparing it with different training objectives. DCN with a single MSE objective function (DCN-MSE), DCN with single objective of multiple mixed terms, the sum of CC and SAM (DCN-SAM-CC), and DCN with multiple objectives, spatial and spectral objectives (MO-DCN), are investigated. For MO-DCN, the spectral objective is considered in the first stage, which is followed by the spatial objective. These objectives are used for training the same network for fair comparison.

The results are shown in Tables XIV–XVI. For DCN-MSE and DCN-SAM-CC, they adopt a single objective function (independent term or multiple mixed terms) to optimize the network, which causes the unbalance performance between spatial and spectral reconstructions according to CC and SAM results of DCN-MSE and DCN-SAM-CC. In contrast to DCN-MSE, MO-DCN achieves excellent comprehensive indices (RMSE, ERGAS) and individual indices (CC, SAM, PSNR) by modeling HP reconstruction along with specific spatial and spectral objectives rather than one integrated objective. Compared with DCN-SAM-CC, MO-DCN obtains an excellent

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TABLE XI
RESULTS OF DIFFERENT NETWORK STRUCTURES ON THE BOTSWANA DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
SPA	0.9187	2.9813	0.0261	2.4694	31.8578
SPA-SPE	0.9204	2.8838	0.0260	2.4174	31.8856
(SPA-SPE-AIA) DCN	**0.9535**	**2.6815**	**0.0235**	**2.0931**	**33.7509**

TABLE XII
RESULTS OF DIFFERENT NETWORK STRUCTURES ON THE CHIKUSEI DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
SPA	0.8874	4.8930	0.0304	6.8807	30.3538
SPA-SPE	0.9105	3.6057	0.0282	4.8155	31.0322
(SPA-SPE-AIA) DCN	**0.9417**	**3.4151**	**0.0234**	**4.3033**	**32.9229**

TABLE XIII
RESULTS OF DIFFERENT NETWORK STRUCTURES ON THE PAVIA CENTER DATASET

Quantitative indices	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	PSNR (↑)
SPA	0.9326	8.2129	0.0318	6.5789	30.4282
SPA-SPE	0.9549	5.2577	0.0272	5.6128	31.6840
(SPA-SPE-AIA) DCN	**0.9726**	**5.0858**	**0.0229**	**5.0437**	**33.4961**

TABLE XIV
RESULTS OF OPTIMIZATION WITH DIFFERENT TRAINING OBJECTIVES ON THE BOTSWANA DATASET

Indices	DCN-MSE	DCN-SAM-CC	MO-DCN	
			Stage 1: SAM	Stage 2: CC
CC (↑)	0.9140	0.9171	0.9221	**0.9535**
SAM (↓)	2.9693	3.0456	**2.5304**	2.6815
RMSE (↓)	0.0269	0.0271	0.0251	**0.0235**
ERGAS (↓)	2.5328	2.5393	2.3324	**2.0931**
PSNR (↑)	31.6231	31.5132	32.5201	**33.7509**

TABLE XV
RESULTS OF OPTIMIZATION WITH DIFFERENT TRAINING OBJECTIVES ON THE CHIKUSEI DATASET

Indices	DCN-MSE	DCN-SAM-CC	MO-DCN	
			Stage 1: SAM	Stage 2: CC
CC (↑)	0.9261	0.9190	0.9168	**0.9417**
SAM (↓)	4.7479	4.3180	**3.3041**	3.4151
RMSE (↓)	0.0270	0.0273	0.0287	**0.0234**
ERGAS (↓)	5.4508	5.2741	4.5758	**4.3033**
PSNR (↑)	31.3933	31.3069	30.8429	**32.9229**

tradeoff solution among spatial and spectral objectives and best performance in each index, which demonstrates that the Epsilon-constraint-based multiobjective optimization can co-improve the SAM and CC by adding a SAM constraint. For MO-DCN, it first focuses more on spectral preservation. Then in the next stage, it can be seen that the SAM drops slightly and the CC rises significantly, thanks to the SAM constraint. The above process further indicates the effectiveness of the proposed Epsilon-constraint-based multiobjective optimization.

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TABLE XVI
RESULTS OF OPTIMIZATION WITH DIFFERENT TRAINING OBJECTIVES ON THE PAVIA CENTER DATASET

Indices	DCN-MSE	DCN-SAM-CC	MO-DCN	
			Stage 1: SAM	Stage 2: CC
CC (↑)	0.9471	0.9221	0.9160	**0.9726**
SAM (↓)	6.0256	5.6770	**4.7480**	5.0858
RMSE (↓)	0.0297	0.0320	0.0271	**0.0229**
ERGAS (↓)	7.7816	7.1312	5.4050	**5.0437**
PSNR (↑)	30.9102	30.3835	31.3935	**33.4961**

V. CONCLUSION

In this article, a novel MO-DCN method is devised to deal with HP with different optimization objectives. MO-DCN combines bidirectional RNN structure and deconvolution to deal with the high-dimensional data upsampling, which merges the bidirectional long-term spectral extraction into the upsampling procedure to achieve less spectral distortion. In the spatial–spectral fusion, the DCN is customized to deal with the distinctive spatial and spectral reconstructions, respectively, and provide more potential for performance improvement. The AIAM of the DCN exploits the complementary spatial information from LR-HSI and PANI and refines the attention in a coarse-to-fine way to realize extraordinary spatial enhancement. These designs work together to form a spatial–spectral fusion uniform framework. Finally, the spatial–spectral fusion is formulated as Epsilon-constraint-based multiobjective optimization. It dynamically fuses PANI and LR-HSI by transforming the conflict between spatial and spectral objectives to a constraint for co-optimization, which meets the spatial–spectral preservation requirements. The design of network architecture is an essential issue in deep learning and it can be well-addressed by multiobjective optimization. In the further, we look forward to combining the multiobjective architecture learning and HP into a uniform framework to gain better potential performance improvement.

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